

Lotka-Volterra modeling three species interaction using the `gauseR` package

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1 Abstract

2 Introduction

- Introduction (mainly include new facts that you needed to learn for this project, mainly exclude general background)

2.1 Aim and project overview

This is a project for the “Practical Course in Modelling and Systems Biology” (2026S 300332-1). The aim of the project is to implement a Lotka-Volterra model using the `gauseR` to model a three species interaction experiment in spatial microbiology.

The experiment consists of a co-culturing experiment on microfluidic chips comprised with micro-scale mazes of different fractal complexity. The species used for co-culturing the two bacterial species *Pseudomonas putida* and *Bacillus subtilis* and the fungal species *Trichoderma* *ssp* (?). All species are commonly found and co-isolated from soil habitats and are therefore thought to be coexisting in those ecosystems.

I originally wanted to conduct the modeling analyses with my own data from co-culturing experiments with *P. putida*, *B. subtilis* and a complex soil bacterial community isolated from soil samples as a natural inoculum. However, since I am not having the necessary experimental data ready in time for the due date of this report I herein use the data of my colleague who conducted similar experiments with *Trichoderma* as the third species.

The overall aim is to see whether the experimental data can be properly represented by a Lotka-Volterra model and investigate if that approach is a viable method to narrow in on mechanistic insights to the interactions of this three species model in a spatially structured habitat.

From the literature it is known, that P.p. and B.s. are frequently co-isolated from soils but cannot coexist in a well mixed environment under lab conditions. Prior experiments (not published) on microfluidic chips have shown that the introduction of a spatial structure allows for stable coexistence of these two species by niche differentiation.

The underlying question of these experiments specifically are if *Trichoderma* alters the interactions between P.p and B.s. facilitating or hindering coexistence, by suppression or support of one or either of the species.

On the broader scope of theoretical ecology the objective is to understand how community composition and (spatial) habitat complexity possibly facilitates coexistence allowing for the high diversity we can observe in nature. Micro-scale spatial structure is a key part of this framework since it is reducing interactions between (microbial) species (and organisms) and

can thereby theoretically infinitely create niche differentiation overcoming potential fitness differences that would otherwise prohibit coexistence (Orr et al. 2025)

In this setup we are tackling two shortcomings of most experimental microbial ecology research: Most research is focused on two species interactions exclusively but it is unclear how these results translate into more complex communities, since higher order interactions might be leading to emergent properties of microbial communities. Secondly, most experiments are carried out in well-mixed liquid cultures, omitting the importance of spatial structure as outlined above.

Underrepresentation of these approaches might be one explanation to the unculturability problem that microbiological research faces since it's very beginnings.

With the approach to reflect the interaction dynamics of this three species system in a simple Lotka-Volterra model I am trying to approximate on the dynamics with a simple model approach to see whether experimental data can be fitted satisfyingly on this degree of complexity already or if more sophisticated models need to be employed. Also the attempt to model the data with a LV model can give insights if the interactions can be described by transient competition or if higher-order interactions are at work.

The Lotka-Volterra equation is while being a mathematically relatively straight forward model still a very powerful approach in ecology that can account for very complex and multi-layered network of interactions without the need to have a complete mechanistic understanding of these interactions on a molecular level.

This is the basic equation we are going to assume as Lotka-Volterra equation throughout is derived from Mühlbauer et al. (2020):

Derived from Mühlbauer et al. (2020) the basic Lotka-Volterra equation we are employing throughout this study describes the changes per unit time in the abundance of species N_i as

$$\frac{dN_i}{dt} = N_i(r_i + a_{ii}N_i + \sum_j (a_{ij}N_j)) \quad (1)$$

where r_i is the intrinsic growth rate... This is a reference to [?@fig-gfp-f5](#) which would be inline or in parentheses ([?@fig-gfp-f5](#)). Moreover I already explained Equation 1 before.

Generalized LV:

$$\frac{dx_i(\mathbf{t})}{d\mathbf{t}} = \mathbf{x}_i(\mathbf{t}) \left(\mathbf{b}_i + \sum_{j=1}^n \mathbf{a}_{ij}x_j(\mathbf{t}) \right) \quad i = 1, \dots, n$$

3 Methods and Data

- Methods and data (provides reproducibility:

Table 1: Estimated Lotka-Volterra interaction coefficients for cocultures of *P. putida* (1) and *B. subtilis* (2) on microfluidic chips with mazes of increasing fractal complexity (f0-f5). a_{11} and a_{22} are the intra-specific effects of *P. putida* and *B. subtilis*, respectively; a_{12} and a_{21} are the inter-specific effects of *B. subtilis* on *P. putida* and vice versa. Negative coefficients indicate competition/inhibition, positive coefficients promotion/enhancement.

Fractal	<i>P. putida</i>	<i>B. subtilis</i>	<i>P. putida</i> + <i>B. subtilis</i>				<i>T (name?)</i> + <i>P. putida</i>	<i>T (name?)</i> + <i>B. subtilis</i>	<i>T (name?)</i> + <i>P. putida</i> + <i>B. subtilis</i>			
	a_{11}	a_{22}	a_{11}	a_{22}	a_{12}	a_{21}	a_{11}	a_{22}	a_{11}	a_{22}	a_{12}	a_{21}
f0	-5e-04	0.0000	-5e-04	-1e-04	0.0014	1e-04	-3e-04	0e+00	-1e-04	-2e-04	-5e-04	0
f1	-7e-04	0.0000	-3e-04	-1e-04	0.0009	1e-04	-2e-04	1e-04	-2e-04	-1e-04	-2e-04	0
f3	-9e-04	-0.0181	-1e-04	-4e-04	0.0003	1e-04	-3e-04	0e+00	-2e-04	-1e-04	-4e-04	0
f5	-7e-04	-0.0082	-6e-04	0e+00	0.0015	0e+00	-3e-04	0e+00	-4e-04	-1e-04	9e-04	0

3.1 Experimental methods

Very brief and general.

3.2 Software and Code

- data incl accession numbers+citation,
- software incl version+non-default parameters+citation,
- databases incl release number or release date + citation,

3.3 Data analysis and modeling workflow

- your workflows; BUT: usually no command lines

4 Results

Here begins the text describing the results

First figure:

Second figure:

The table follows here.

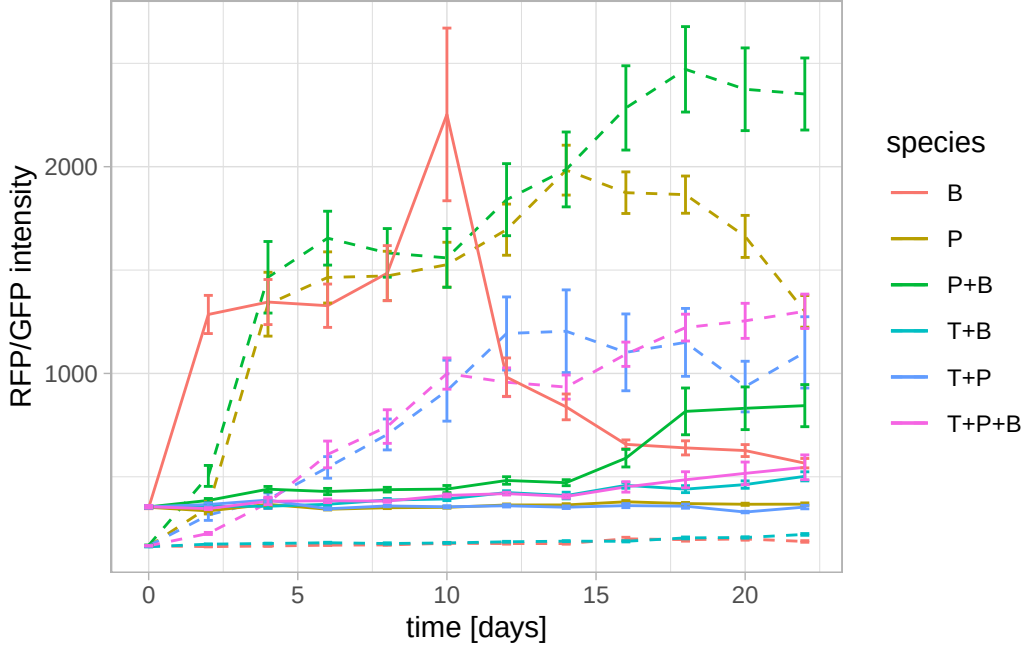


Figure 1: RFP (solid line) and GFP (dashed line) fluorescence intensity on microfluidic mazes with fractal order 5.

4.1 Heatmap of interaction coefficients

4.2 Lotka—Volterra modeling using the `gauseR` library

4.2.1 Parameterization and regression fit for *Pseudomonas* monoculture

Time-lagged data was generated using the `get_lag` function and the obtained data was used to calculate the per-capita growth rate.

The per-capita growth rate for the logistic growth model was defined as:

$$\frac{dN}{dt}/N = r - a_{ii}N$$

with r being the intrinsic growth rate, and a_{ii} the self-limitation in growth of the species.

The per-capita growth rate was plotted against the abundance and a linear regression was fitted to obtain the needed parameters r (read as the y-intercept) and K (read as the x-intercept). a_{ii} is the slope of the linear regression.

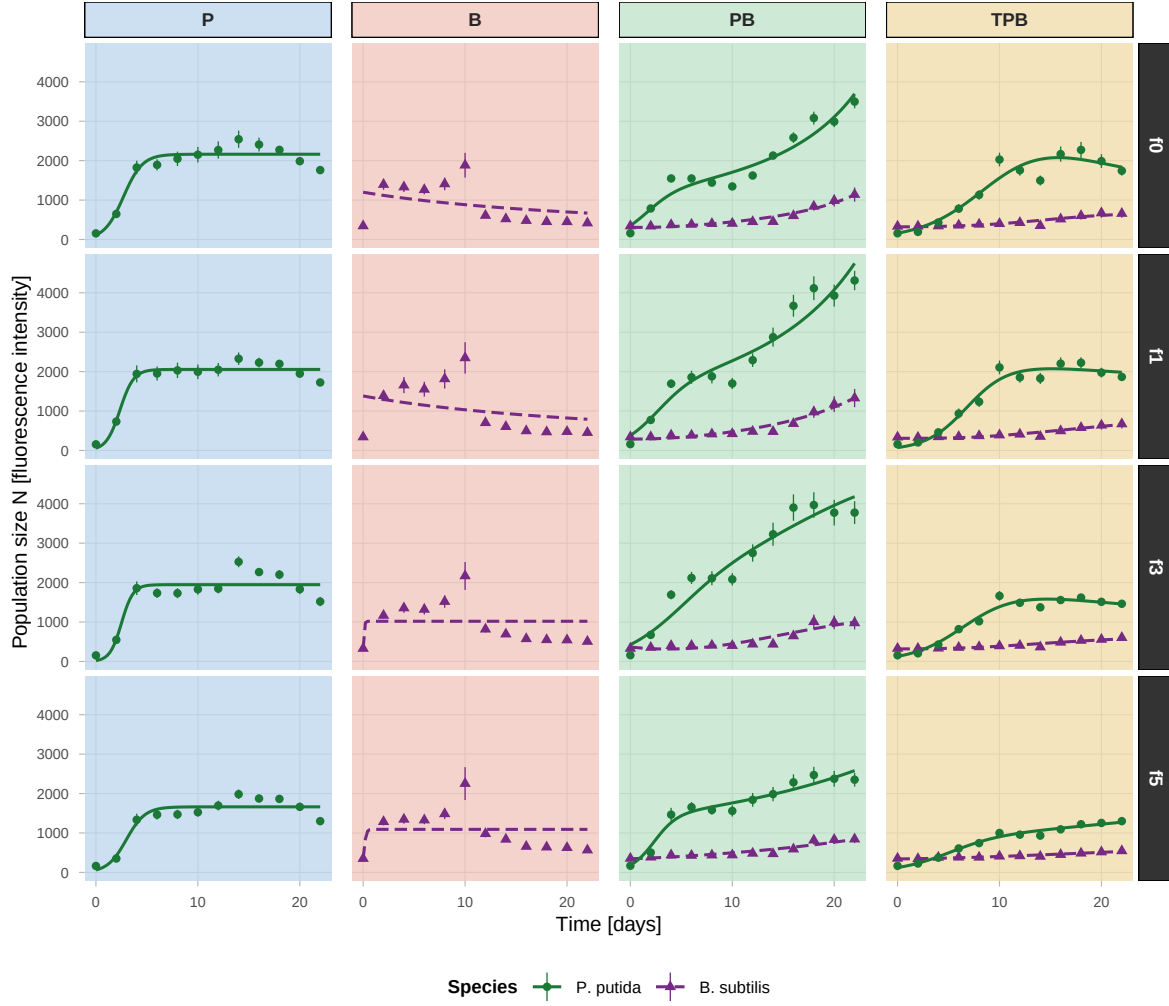


Figure 2: Observed (points, $\pm$ SE) and predicted (lines) growth curves across fractal levels (rows) and conditions (columns).

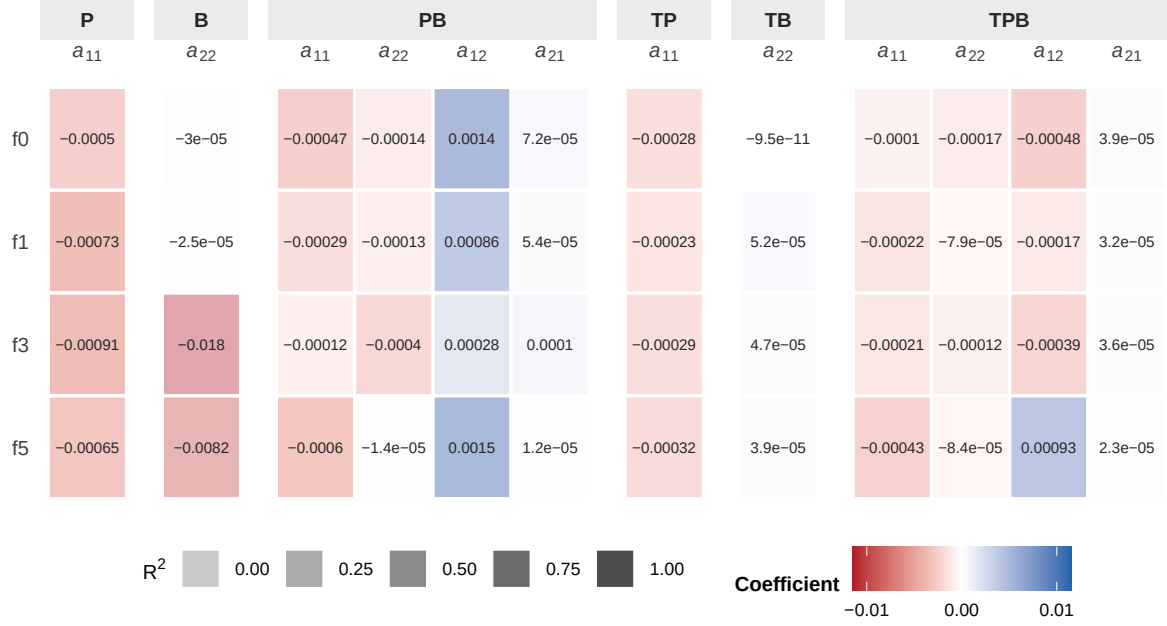


Figure 3: Interaction coefficients per fractal level (rows) and condition (columns). Tile colour shows the estimated coefficient (blue: positive, red: negative); colour saturation shows R^2 of the fit of the species whose growth equation contains the coefficient (faint = poor fit). Numbers are the coefficient values.

5 Discussion

- Discussion (did we succeed w.r.t. the project aims? did we make particular assumptions? do our methods have weaknesses? how do the results compare to previous expectations and previous work)

6 Statement of authorship and use of AI and LLMs

This report is submitted as the practical project for the class “Practical Course in Modelling and Systems Biology” (2026S 300332-1). The used experimental data has been gathered and provided by Miryam Al-Jalahwee, BSc. All data arranging, sorting and subsequent analysis has been conducted by the author.

7 References

- Mühlbauer, Lina K., Maximilienne Schulze, W. Stanley Harpole, and Adam T. Clark. 2020. “gauseR: Simple Methods for Fitting Lotka-Volterra Models Describing Gause’s ‘Struggle for Existence’” *Ecology and Evolution* 10 (23): 13275–83. <https://doi.org/10.1002/ece3.6926>.
- Orr, James A., David W. Armitage, and Andrew D. Letten. 2025. “Coexistence Theory for Microbial Ecology, and Vice Versa.” *Environmental Microbiology* 27 (3): e70072. <https://doi.org/10.1111/1462-2920.70072>.

8 Supplementary Material

- Supplementary material: code and command lines (if needed)

Table 2: Goodness of fit of the fitted growth curves against the observed means, per species. $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ (can be negative for fits worse than the mean); RMSE in the units of N (fluorescence intensity). Model predictions were interpolated at the observed time points.

Condition	Fractal	<i>P. putida</i>		<i>B. subtilis</i>	
		R^2	RMSE	R^2	RMSE
<i>P. putida</i>	f0	0.913	201.964	—	—
	f1	0.942	150.121	—	—
	f3	0.836	263.606	—	—
	f5	0.885	186.036	—	—
<i>B. subtilis</i>	f0	—	—	0.169	466.131
	f1	—	—	0.137	608.991
	f3	—	—	0.133	487.614
	f5	—	—	0.161	467.490
<i>P. putida</i> + <i>B. subtilis</i>	f0	0.935	240.730	0.956	55.194
	f1	0.929	343.371	0.949	75.043
	f3	0.930	323.308	0.861	95.776
	f5	0.954	148.544	0.880	60.509
<i>T (name?)</i> + <i>P. putida</i>	f0	0.966	99.799	—	—
	f1	0.969	124.024	—	—
	f3	0.942	126.850	—	—
	f5	0.936	90.757	—	—
<i>T (name?)</i> + <i>B. subtilis</i>	f0	—	—	0.845	22.368
	f1	—	—	0.951	13.095
	f3	—	—	0.956	10.777
	f5	—	—	0.943	11.091
<i>T (name?)</i> + <i>P. putida</i> + <i>B. subtilis</i>	f0	0.909	225.997	0.865	45.482
	f1	0.960	151.623	0.896	38.554
	f3	0.959	110.466	0.904	28.892
	f5	0.979	54.865	0.933	15.762

Condition	Fractal	Parameter	Estimate
<i>P. putida</i>	f0	P_0	114.2127
		r_1	1.0713
		a_{11}	-0.0005
	f1	P_0	58.4481
		r_1	1.4975
		a_{11}	-0.0007
	f3	P_0	22.7074
		r_1	1.7716
		a_{11}	-0.0009
	f5	P_0	62.4204
		r_1	1.0877
		a_{11}	-0.0007
<i>B. subtilis</i>	f0	B_0	1199.2839
		r_2	0.0000
		a_{22}	-0.0000
	f1	B_0	1382.1754
		r_2	0.0000
		a_{22}	-0.0000
	f3	B_0	327.2705
		r_2	18.3931
		a_{22}	-0.0181
	f5	B_0	350.1735
		r_2	8.9568
		a_{22}	-0.0082

41007933459505153095614286719776033139076752342243348972
37577128290565751666608145976294101544199978015460422

(continued)

Condition	Fractal	Parameter	Estimate
<i>P. putida</i> + <i>B. subtilis</i>	f0	B_0	310.5821
		P_0	353.4461
		r_1	0.2093
		r_2	0.0020
		a_{11}	-0.0005
		a_{22}	-0.0001
	f1	a_{12}	0.0014
		a_{21}	0.0001
		B_0	289.3245
		P_0	381.6527
		r_1	0.2973
		r_2	0.0165
	f3	a_{11}	-0.0003
		a_{22}	-0.0001
		a_{12}	0.0009
		a_{21}	0.0001
		B_0	367.1730
		P_0	425.9024
	f5	r_1	0.2497
		r_2	0.0189
		a_{11}	-0.0001
		a_{22}	-0.0004
		a_{12}	0.0003
		a_{21}	0.0001
<i>T</i> (name?) + <i>P. putida</i>	f0	B_0	346.1842
		P_0	185.5721
		r_1	0.3586
	f1	r_2	0.0271
		a_{11}	-0.0006
		a_{22}	-0.0000
	f3	a_{12}	0.0015
		a_{21}	0.0000
	f5	B_0	97.2307
		r_1	0.4414
		a_{11}	-0.0003
	f1	P_0	84.7892
		r_1	0.4556
		a_{11}	-0.0002
	f3	P_0	93.9718
		r_1	0.4456
		a_{11}	-0.0003
	f5	P_0	129.0483
		r_1	0.3608
		a_{11}	-0.0003

(continued)

Condition	Fractal	Parameter	Estimate
<i>T</i> (name?) + <i>B. subtilis</i>	f0	B_0	314.1291
		r_2	0.0194
		a_{22}	-0.0000
	f1	B_0	322.0550
		r_2	0.0000
		a_{22}	0.0001
	f3	B_0	322.1593
		r_2	0.0000
		a_{22}	0.0000
	f5	B_0	344.2782
		r_2	0.0000
		a_{22}	0.0000
<i>T</i> (name?) + <i>P. putida</i> + <i>B. subtilis</i>	f0	B_0	320.5456
		P_0	160.2818
		r_1	0.4678
		r_2	0.0520
		a_{11}	-0.0001
		a_{22}	-0.0002
		a_{12}	-0.0005
	f1	a_{21}	0.0000
		B_0	308.3955
		P_0	72.9189
		r_1	0.5479
		r_2	0.0213
		a_{11}	-0.0002
		a_{22}	-0.0001
	f3	a_{12}	-0.0002
		a_{21}	0.0000
		B_0	318.1260
		P_0	128.4779
		r_1	0.5132
		r_2	0.0355
		a_{11}	-0.0002
	f5	a_{22}	-0.0001
		a_{12}	-0.0004
		a_{21}	0.0000
		B_0	341.1477
		P_0	122.7393
		r_1	0.0742
		r_2	0.0364
		a_{11}	-0.0004
		a_{22}	-0.0001
		a_{12}	0.0009
		a_{21}	0.0000